

Sharp Conditioning Laws for Uniform Random Series

Variance thresholds, boundary phases,
Brownian bridges, and rare-event sampling

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28 September 2026

Abstract

Let $S = \sum_{k \geq 1} w_k U_k$, where the U_k are independent uniform random variables and the positive weights are summable. What does an entire configuration look like when S is forced close to zero? We prove a conditional-law theorem for the moving exponential tilt of this series. For every positive summable weight sequence, a coordinate block is asymptotically unaffected by the conditioning in total variation if and only if it carries a vanishing fraction of the tilted variance. No monotonicity or regular spacing of the weights is required. For a limiting variance fraction $\alpha \in (0, 1)$, we calculate the exact nonzero total-variation profile and the relative-entropy limit $[-\log(1 - \alpha) - \alpha]/2$. At fraction one, total variation tends to one and relative entropy diverges.

For geometric weights, including the classical Fabius distribution, we identify an infinite, phase-dependent boundary product law, prove a Brownian-bridge limit for conditioned partial sums, and show that the rescaled gap to the conditioning boundary becomes an independent exponential random variable. The same variance criterion applies to stretched-exponential and polynomial weights. A lazy rejection sampler is exact in a real-arithmetic model and uses asymptotically $\sqrt{2\pi} n^{3/2}$ scalar uniform draws per accepted geometric configuration, where n is the effective dimension.

The proofs are given in full. Exact finite-simplex formulas and reproducible numerical experiments check the constants, but are not used as substitutes for proofs. The contribution is a conditional configuration theory connected to the repository's scalar Fabius endpoint program; neither global priority nor Lean verification is asserted.

Scope and status. Independent research draft, not externally refereed. Exponential tilting and the Gibbs conditioning principle are classical. The theorem package below is proved here under its displayed hypotheses; its precise relationship to the full prior literature remains a priority-review task. Repository snapshot: e2b1f016a94102663f12b970e6dd434229f90d01.

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1 From scalar endpoint probabilities to conditioned configurations

1.1 The connection with the repository

The Fabius–Rvachev development in `ProveIt` contains a substantial endpoint-asymptotic program. The pinned source `PaperFabiusAsymptotic.lean` distinguishes the valid Laplace-product and saddle-point route from an older invalid residual argument, and records all-orders exact-phase results and carefully scoped inverse statements [7]. The present article starts with the same probabilistic object but asks a different question: not merely *how small is the endpoint probability*, but *which configurations realize that event*.

The original probabilistic construction of Fabius and the later exposition of Arias de Reyna supply the historical context [1, 2]. In the normalization used here,

$$S_{1/2} = \sum_{k \geq 1} 2^{-k} U_k, \quad F(x) = P(S_{1/2} \leq x), \quad U_k \sim \text{Unif}[0, 1]. \quad (1.1)$$

Thus F is the bounded distribution function on $[0, 1]$, not a signed extension of the Fabius function outside that interval. The general geometric model is $S_q = \sum q^k U_k$, $0 < q < 1$.

Four precise research questions organize this article.

Q1: Independence threshold.

How large can a collection of coordinates be before the constraint $S \leq x$ changes its product-tilted law by a nonvanishing amount?

Q2: The price of a macroscopic block.

When that change does not vanish, can one calculate its exact information-theoretic size, rather than only an upper bound?

Q3: Bulk and boundary geometry.

What are the fluctuation process, the distance to the constraint, and the transition between strongly tilted and almost untouched coordinates?

Q4: Constructive realization.

Can one sample the conditioned law without waiting for an event whose original probability is extremely small?

These are questions formulated here from the repository’s endpoint program. We do not identify them with a named published conjecture, or claim that a keyword search establishes their absence from every repository manuscript.

1.2 Prior work and the novelty boundary

Conditional approximation by exponential families is classical. Diaconis and Freedman developed conditional limit theorems and finite forms of de Finetti’s theorem [3]; Dembo and Zeitouni explicitly considered growing blocks and sharp Gibbs-conditioning bounds [4]. Consequently, the assertion that small blocks approach a tilted product law is *not* presented as a new principle.

A recent comparison point is Cator and Ferrari’s Gibbs principle for independent, not necessarily identically distributed, log-concave integer-valued variables [5]. Their setup uses a fixed tilt and a growing finite sum, with a no-condensation condition. Here we work with a continuous infinite weighted series, let the tilt tend to infinity, and compute marginal total-variation and entropy profiles for an arbitrary deterministic subset of coordinates. Our proofs do not invoke their theorem. Their result is important context, not a missing lemma being silently assumed.

The substantive development below is the combination of an explicit variance-fraction theorem for every positive summable weight sequence, its complete endpoint dichotomy, an infinite geometric boundary law, a joint bridge–slack limit, and a proved lazy-sampling cost. The Gaussian profile itself is naturally related to classical ensemble equivalence and already appears in a finite-simplex calculation. What is established here is its validity for the specified moving-tilt random-series problem, with the hypotheses and consequences made explicit. The removal of weight regularity is obtained by using log-concavity to upgrade a central limit theorem to uniform density convergence. A targeted literature check is not a certification of worldwide novelty.

1.3 Conventions

All random coordinates are indexed by $\mathbb{N} = \{1, 2, \dots\}$. All logarithms are natural. For probability measures P, Q on the same measurable space,

$$d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2} \int |dP - dQ|, \quad D_{\text{KL}}(P\|Q) = \int \log \frac{dP}{dQ} dP,$$

with relative entropy $+\infty$ when absolute continuity fails. The standard normal density and distribution function are φ and Φ . An exponential random variable has rate one unless stated otherwise. The notation $o(1)$ always refers to $t \rightarrow \infty$, except where a different limiting variable is specified.

2 The variance-fraction theorem

Let $w_k > 0$ and $\sum_k w_k < \infty$. On the product uniform probability space (Ω, P) define

$$S = \sum_{k \geq 1} w_k U_k, \quad L(t) = \mathbb{E}_P e^{-tS}, \quad \frac{dQ_t}{dP} = \frac{e^{-tS}}{L(t)}, \quad t > 0.$$

Set $a_k = t w_k$ and introduce the functions

$$\mu(a) = 1 - \frac{a}{e^a - 1}, \quad v(a) = 1 - \frac{a^2 e^a}{(e^a - 1)^2}, \quad a > 0. \quad (2.1)$$

The tilted coordinates $Y_k = a_k U_k$ are independent; their means and variances are $\mu(a_k)$ and $v(a_k)$. Define

$$m = m(t) = \sum_k \mu(a_k), \quad V = V(t) = \sum_k v(a_k), \quad x = x(t) = \frac{m(t)}{t}. \quad (2.2)$$

Finally let

$$P_t = P(\cdot \mid S \leq x(t)), \quad D_t = t(x(t) - S) \quad \text{under } P_t. \quad (2.3)$$

The map $t \mapsto x(t)$ parametrizes every sufficiently small positive endpoint; this is proved in Proposition 3.1.

No monotonicity, regular variation, geometric structure, or nonconcentration hypothesis is imposed on the weights. Summability and positivity are enough. The local density theorem that makes this possible is proved in Section 4.

For a deterministic subset $I = I_t \subseteq \mathbb{N}$, possibly infinite, put

$$V_I = \sum_{k \in I} v(a_k), \quad \alpha_t = V_I/V, \quad P_{t,I} = \mathcal{L}_{P_t}((U_k)_{k \in I}), \quad Q_{t,I} = \mathcal{L}_{Q_t}((U_k)_{k \in I}).$$

There is no requirement that I_t be an initial segment, nested, or chosen by a fixed rule independent of t . It must not depend on the realized random coordinates.

Theorem 2.1 (Sharp variance threshold and information profiles). *For every positive summable weight sequence, if $\alpha_t \rightarrow \alpha \in [0, 1)$, then*

$$d_{\text{TV}}(P_{t,I}, Q_{t,I}) \rightarrow \Delta(\alpha), \quad (2.4)$$

$$D_{\text{KL}}(P_{t,I} \| Q_{t,I}) \rightarrow K(\alpha) := \frac{-\log(1-\alpha) - \alpha}{2}. \quad (2.5)$$

Here $\Delta(0) = 0$, and for $0 < \alpha < 1$,

$$\Delta(\alpha) = 2 \left[\Phi(d_\alpha) - \Phi(\sqrt{1-\alpha} d_\alpha) \right], \quad d_\alpha = \sqrt{\frac{-\log(1-\alpha)}{\alpha}}. \quad (2.6)$$

Moreover,

$$d_{\text{TV}}(\mathcal{L}_{P_t}((U_k)_{k \in I}, D_t), P_{t,I} \otimes \text{Exp}(1)) \rightarrow 0. \quad (2.7)$$

If $\alpha_t \rightarrow 1$, then total variation tends to 1 and relative entropy tends to $+\infty$.

Without assuming that α_t converges, the following three conditions are equivalent:

$$V_I = o(V), \quad d_{\text{TV}}(P_{t,I}, Q_{t,I}) \rightarrow 0, \quad D_{\text{KL}}(P_{t,I} \| Q_{t,I}) \rightarrow 0. \quad (2.8)$$

The quantity governing independence is therefore *variance*, not the number of selected coordinates. In a geometric series most active coordinates have variance close to one, so count and variance agree to leading order. For polynomial weights they do not agree with the same proportionality throughout the active range.

The profile has the endpoint behavior

$$\Delta(\alpha) \sim \varphi(1)\alpha, \quad K(\alpha) \sim \frac{\alpha^2}{4} \quad (\alpha \downarrow 0), \quad \Delta(\alpha) \rightarrow 1 \quad (\alpha \uparrow 1). \quad (2.9)$$

For example, a block carrying half the variance has limiting total variation approximately 0.1660641 and relative entropy 0.0965736 nats. Such a block is not asymptotically independent of the constraint, even though every fixed collection inside it can have the usual product limit.

3 The tilted coordinates and the endpoint parameter

Proposition 3.1 (Exact product change of measure). *For every $t > 0$,*

$$L(t) = \prod_{k \geq 1} \frac{1 - e^{-t w_k}}{t w_k} > 0. \quad (3.1)$$

Under Q_t , the coordinates $Y_k = t w_k U_k$ are independent with density

$$h_a(y) = \frac{e^{-y}}{1 - e^{-a}} \mathbf{1}_{[0,a]}(y). \quad (3.2)$$

The sums in Equation (2.2) are finite, $V(t) \rightarrow \infty$, and

$$x'(t) = -V(t)/t^2 < 0, \quad x(0+) = \frac{1}{2} \sum_k w_k, \quad x(\infty) = 0.$$

In particular $x(t)$ is a continuous bijection from $(0, \infty)$ onto $(0, \frac{1}{2} \sum_k w_k)$.

Proof. Since $0 \leq S \leq \sum w_k$, bounded convergence gives the product formula from finite products. Its positivity also follows immediately from $e^{-tS} \geq e^{-t \sum w_k} > 0$. Factoring the exponential Radon–Nikodym derivative on every cylinder proves independence and Equation (3.2); cylinder identities determine the countable product law.

Direct integration by parts shows that the mean and variance of Equation (3.2) are Equation (2.1). They satisfy $0 < \mu(a) \leq a/2$ and $0 < v(a) < 1$. The first inequality follows because an exponentially decreasing weight on $[0, a]$ moves the uniform mean downward. Also $v(a) \leq a^2/4$, the elementary range bound for variance. Thus $m(t) \leq (t/2) \sum w_k$ and $V(t) \leq (t^2/4) \sum w_k^2 < \infty$. For any fixed M , $\sum_{k \leq M} v(tw_k) \rightarrow M$; hence $V(t) \rightarrow \infty$.

Differentiation under expectation is valid because S is bounded. It yields $x(t) = -(\log L)'(t) = \mathbb{E}_{Q_t} S$ and $x'(t) = -\text{Var}_{Q_t} S = -V(t)/t^2$. The limiting value at zero follows from dominated convergence. For each fixed k , the tilted mean of $w_k U_k$ is $\mu(tw_k)/t \rightarrow 0$, and it is bounded by $w_k/2$. Dominated convergence for the summable series proves the limit at infinity. Strict monotonicity and continuity finish the proof. $\square$

Lemma 3.2 (Uniform moment estimates). *For every real $p \geq 2$ there is a finite constant C_p such that, for H_a of density Equation (3.2),*

$$\mathbb{E}|H_a - \mu(a)|^p \leq C_p v(a), \quad a > 0. \quad (3.3)$$

There are absolute constants $c, C > 0$ for which

$$c \min(a^2, 1) \leq v(a) \leq C \min(a^2, 1), \quad a > 0.$$

The small-width expansions are

$$\mu(a) = \frac{a}{2} - \frac{a^2}{12} + \frac{a^4}{720} - \frac{a^6}{30240} + O(a^8), \quad (3.4)$$

$$v(a) = \frac{a^2}{12} - \frac{a^4}{240} + \frac{a^6}{6048} + O(a^8), \quad (3.5)$$

and $v(a) = \mu(a) - a\mu'(a)$.

Proof. The variable H_a/a has density $ae^{-au}/(1 - e^{-a})$ on $[0, 1]$, converging uniformly to one as $a \downarrow 0$. Its centered p th absolute moment stays bounded, while its variance tends to $1/12$. Thus the ratio in Equation (3.3) is $O(a^{p-2})$ at zero. As $a \rightarrow \infty$, the densities h_a converge to the exponential density and, for $a \geq 1$, are bounded by $(1 - e^{-1})^{-1}e^{-y}$ on $[0, \infty)$. All relevant moments converge; in particular $v(a) \rightarrow 1$. On each remaining compact interval of positive widths the ratio is continuous with a positive denominator. These three regions prove the assertions about moments and variance. Taylor expansion of $a/(e^a - 1)$ gives Equations (3.4) and (3.5); direct differentiation gives the last identity. $\square$

For a subset $J \subseteq \mathbb{N}$, write

$$Z_J = \sum_{k \in J} (Y_k - \mu(a_k)), \quad V_J = \text{Var}_{Q_t}(Z_J). \quad (3.6)$$

The sum converges absolutely, since $\sum |Y_k - \mu(a_k)| \leq \sum (a_k + \mu(a_k)) < \infty$ for fixed t . This elementary fact avoids any ambiguity about rearranging or splitting infinitely many coordinates.

4 A local central limit theorem without weight regularity

The crucial analytic point is a *density* approximation, not merely a central limit theorem. The conditioning is sensitive to a window of width one in Z , whereas its standard deviation is $\sqrt{V}$. A Fourier tail estimate based on many support widths bounded away from zero would introduce an unnecessary restriction. Log-concavity removes that restriction entirely.

4.1 Why every subset sum has a log-concave density

A nonnegative function f is log-concave if

$$f((1 - \lambda)x + \lambda y) \geq f(x)^{1-\lambda} f(y)^\lambda \quad (0 < \lambda < 1).$$

Zeros outside an interval are allowed. The densities h_a in Equation (3.2) are log-concave. We include the required convolution argument, a one-dimensional instance of the classical Prékopa–Leindler mechanism [6].

Lemma 4.1 (Convolution and infinite sums). *The convolution of two integrable log-concave functions on $\mathbb{R}$ is log-concave. For fixed t and any subset J containing at least two indices, Z_J has a bounded continuous compactly supported log-concave density.*

Proof. First establish the integral inequality needed for convolution. Suppose F, G are positive and continuous on the interiors of their interval supports, have finite positive integrals, and

$$H((1 - \lambda)x + \lambda y) \geq F(x)^{1-\lambda} G(y)^\lambda.$$

Normalize F, G to have integral one, rescaling H accordingly. Let T be the increasing transport matching their distribution functions. On the interiors, T is continuously differentiable and $G(T(x))T'(x) = F(x)$. Set $z(x) = (1 - \lambda)x + \lambda T(x)$. Its derivative is positive. Change variables over its image and use the displayed hypothesis to obtain

$$\begin{aligned} \int H(z) dz &\geq \int F(x)^{1-\lambda} G(T(x))^\lambda [(1 - \lambda) + \lambda T'(x)] dx \\ &= \int F(x) [(1 - \lambda)T'(x)^{-\lambda} + \lambda T'(x)^{1-\lambda}] dx \geq \int F(x) dx = 1. \end{aligned}$$

The last inequality is the weighted arithmetic–geometric mean inequality. Endpoint values have no effect on these integrals; improper endpoints are handled by increasing limits. Undoing the normalization proves $\int H \geq (\int F)^{1-\lambda} (\int G)^\lambda$.

For log-concave f, g , fixed z_0, z_1 , and $z_\lambda = (1 - \lambda)z_0 + \lambda z_1$, apply this inequality to

$$F(x) = f(x)g(z_0 - x), \quad G(y) = f(y)g(z_1 - y), \quad H(u) = f(u)g(z_\lambda - u).$$

Log-concavity of f and g verifies the hypothesis. A nonzero integrable log-concave function is positive and continuous on the interior of its interval support, so the preceding proof applies. If an integral vanishes, the desired convolution inequality is trivial. This establishes the convolution assertion in the case needed here and, with the usual immaterial endpoint choices, in general.

For Z_J , finite convolution now proves log-concavity. The convolution of the first two chosen bounded interval densities is continuous, bounded, and compactly supported; call it g . If J is infinite, convolve g with the law of the remaining partial sum. Those partial-sum densities converge uniformly to the full sum density: the omitted uncentered tail is bounded by the sum of its widths, which tends to zero, and g is uniformly continuous. More precisely, their supremum difference is at most the modulus of continuity of g evaluated at that tail bound. The uniform limit is continuous and log-concave. Its support stays inside an interval of length $\sum_{k \in J} a_k < \infty$. Centering is an affine translation and preserves all these properties. $\square$

4.2 A shape-based upgrade from weak to uniform convergence

Lemma 4.2 (Log-concave convergence to a Gaussian). *Let f_j be continuous compactly supported log-concave probability densities on $\mathbb{R}$. If their laws converge weakly to $N(0, 1)$, then*

$$\sup_x |f_j(x) - \varphi(x)| \longrightarrow 0. \tag{4.1}$$

Proof. Choose a mode b_j of each f_j , which exists by compactness. Log-concavity implies that f_j is nondecreasing to the left of b_j and nonincreasing to its right. We first prove $b_j \rightarrow 0$. If a subsequence had $b_j \geq \varepsilon > 0$, its densities would be nondecreasing on $(0, \varepsilon)$. Two disjoint, ordered intervals of the same length inside that interval would then have right-hand mass at least left-hand mass. Weak convergence passes this inequality to the Gaussian, whose density is strictly decreasing there, a contradiction. A subsequence with $b_j \leq -\varepsilon$ is ruled out on the negative side in the same way.

Let F_j be the distribution functions. Weak convergence to a continuous distribution implies uniform convergence $F_j \rightarrow \Phi$: a finite grid controls the compact part by monotonicity, and two endpoint tail bounds control the rest. For x in a compact interval bounded away from zero on the positive side, and a sufficiently small fixed $h > 0$, eventual monotonicity gives

$$\frac{F_j(x+h) - F_j(x)}{h} \leq f_j(x) \leq \frac{F_j(x) - F_j(x-h)}{h}.$$

Uniform convergence of the distribution functions followed by $h \downarrow 0$ proves uniform density convergence on that compact interval. The negative side is identical with reversed inequalities.

It remains to control the peak. Fix $\delta > 0$. Eventually $|b_j| < \delta/2$, and log-concavity at the midpoint δ of b_j and $2\delta - b_j$ gives

$$f_j(b_j) \leq \frac{f_j(\delta)^2}{f_j(2\delta - b_j)}.$$

The denominator is positive eventually, by the convergence already proved away from zero. Hence

$$\limsup_j \|f_j\|_\infty \leq \frac{\varphi(\delta)^2}{\varphi(2\delta)} = \varphi(0)e^{\delta^2}.$$

Let $\delta \downarrow 0$ to get the upper bound $\varphi(0)$. On $[-\delta, \delta]$, unimodality and log-concavity give $f_j(x) \geq \min(f_j(-\delta), f_j(\delta)) \rightarrow \varphi(\delta)$. Together with the peak bound, this proves uniform convergence through zero, by then letting $\delta \downarrow 0$.

Finally, for fixed large R , eventual unimodality implies $\sup_{x \geq R} f_j(x) \leq f_j(R) \rightarrow \varphi(R)$, and similarly on $x \leq -R$. Let $R \rightarrow \infty$. The two tails vanish uniformly, proving Equation (4.1) on the whole real line. $\square$

Theorem 4.3 (Unrestricted subset local central limit theorem). *Let the weights be any positive summable sequence, and let $J = J_t$ be any deterministic subset with $V_J(t) \rightarrow \infty$. Then Z_J has a bounded continuous density $f_{t,J}$ for all sufficiently large t , and*

$$\sup_{z \in \mathbb{R}} \left| \sqrt{V_J} f_{t,J}(z) - \varphi(z/\sqrt{V_J}) \right| \rightarrow 0. \quad (4.2)$$

In particular $\|f_{t,J}\|_\infty \leq C/\sqrt{V_J}$ eventually, and $Z_J/\sqrt{V_J} \Rightarrow N(0, 1)$.

Proof. Put $s = \sqrt{V_J}$. The centered third-moment estimate gives

$$\frac{\sum_{k \in J} \mathbb{E}|Y_k - \mu(a_k)|^3}{s^3} \leq \frac{C_3}{s} \rightarrow 0.$$

Thus the normalized centered sum converges weakly to the standard Gaussian. Here is a direct characteristic-function justification also valid for a countably infinite row. For each fixed u , expand the centered factors at u/s . Their quadratic terms sum to $-u^2/2$ and their cubic remainders have total absolute value at most $C|u|^3/s$. Since $v(a_k) \leq 1$, the sum of the errors in replacing each factor by its logarithm is $O(u^4/s^2)$. Taking the exponential yields convergence of the product to $e^{-u^2/2}$. The same bounds justify the passage from finite to countable products, because all moment sums converge for each fixed t .

Since $V_J \rightarrow \infty$ and each individual variance is at most one, J eventually contains at least two indices. By Lemma 4.1, the normalized sum has a continuous compactly supported log-concave density. Apply Lemma 4.2 along an arbitrary sequence $t \rightarrow \infty$ to obtain Equation (4.2). The sequential criterion for a limit then gives the assertion as stated. Its density bound follows immediately. $\square$

Remark 4.4 (The regularity issue is resolved, not assumed away). An ordinary central limit theorem does not generally imply a local one. Here it does after using the preserved log-concave shape. In particular, a plateau of many very narrow coordinates, an irregular sequence of widely separated scales, or a nonmonotone ordering of the weights does not require a separate smoothing assumption. This is why the main theorem applies to *every* positive summable weight sequence.

5 The boundary layer and the information cost of the full law

Write $Z = Z_N = tS - m$. The exact change of measure gives

$$\frac{dP_t}{dQ_t} = \frac{e^Z \mathbf{1}_{Z \leq 0}}{J(t)}, \quad J(t) = \mathbb{E}_{Q_t}[e^Z \mathbf{1}_{Z \leq 0}]. \quad (5.1)$$

The quantity J is both a saddle normalization and the eventual acceptance probability of our sampler.

Theorem 5.1 (Saddle normalization and exponential slack). *For every positive summable weight sequence,*

$$J(t) \sim \frac{1}{\sqrt{2\pi V(t)}}, \quad (5.2)$$

$$P(S \leq x(t)) \sim \frac{L(t)e^{m(t)}}{\sqrt{2\pi V(t)}}. \quad (5.3)$$

Under P_t , $D_t = -Z$ converges to $\text{Exp}(1)$ in total variation. For every fixed real $r \geq 0$,

$$\mathbb{E}_{P_t} D_t^r \rightarrow \Gamma(r+1). \quad (5.4)$$

Proof. Let $f_t = f_{t,N}$. By Theorem 4.3,

$$J(t) = \int_0^\infty e^{-d} f_t(-d) dd. \quad (5.5)$$

For fixed d , $\sqrt{V} f_t(-d) \rightarrow \varphi(0)$, and $\sqrt{V} f_t(-d) \leq C$ uniformly. Dominated convergence gives Equation (5.2). Equation (5.1) then yields the exact identity $P(S \leq x) = L(t)e^m J(t)$, proving Equation (5.3).

The slack density is

$$p_{D,t}(d) = \frac{e^{-d} f_t(-d)}{J(t)} \mathbf{1}_{[0,\infty)}(d). \quad (5.6)$$

It tends pointwise to e^{-d} and is bounded by Ce^{-d} for all large t . Dominated convergence proves both its L^1 convergence and Equation (5.4). $\square$

Corollary 5.2 (Information and overlap of the full configurations). *For every positive summable weight sequence,*

$$D_{\text{KL}}(P_t \| Q_t) = \frac{1}{2} \log(2\pi V) - 1 + o(1), \quad (5.7)$$

and

$$1 - d_{\text{TV}}(P_t, Q_t) \sim \frac{\log V}{2\sqrt{2\pi V}}. \quad (5.8)$$

Proof. The entropy identity is exact before taking limits:

$$D_{\text{KL}}(P_t \| Q_t) = -\mathbb{E}_{P_t} D_t - \log J(t).$$

Use Equations (5.2) and (5.4). For overlap, put $b = -\log J(t) \sim \frac{1}{2} \log V$. Integration of the minimum of the two densities in Equation (5.1) gives

$$1 - d_{\text{TV}}(P_t, Q_t) = \int_0^b f_t(-d) dd + J(t)^{-1} \int_b^\infty e^{-d} f_t(-d) dd. \quad (5.9)$$

Uniformly on $0 \leq d \leq b$, $d/\sqrt{V} \rightarrow 0$, so the first integral is asymptotic to $b/\sqrt{2\pi V}$ by Equation (4.2). The second is bounded by $(C/\sqrt{V})J^{-1}e^{-b} = C/\sqrt{V}$, which is smaller by a logarithmic factor. This proves Equation (5.8). $\square$

The interpretation is precise: a local block can look independent, while the full configurations become almost mutually singular. The full entropy cost grows only logarithmically with the effective variance, even though total variation tends to its maximum.

6 Proof of the sharp block theorem

6.1 The exact marginal likelihood ratio

Let $I = I_t$, let $J = \mathbb{N} \setminus I$, and assume $V_J \geq \eta V$. By independence under Q_t , the complement has the density supplied by Theorem 4.3. If the realized centered block sum is $z = Z_I$, then the marginal likelihood ratio is

$$R_t(z) = \frac{\int_0^\infty e^{-d} f_{t,J}(-z-d) dd}{J(t)}. \quad (6.1)$$

Indeed, in the numerator of the conditional density one integrates $e^{z+Z_J} \mathbf{1}_{z+Z_J \leq 0}$ and makes the substitution $Z_J = -z-d$. The factors e^z and e^{-z} cancel. This cancellation is responsible for the simple formula.

There is also an exact joint density with respect to $Q_{t,I} \otimes \text{Exp}(1)$:

$$H_t(z, d) = \frac{f_{t,J}(-z-d)}{J(t)}, \quad d \geq 0. \quad (6.2)$$

These identities hold on countable product spaces, including when I is infinite, because they follow from conditioning on the coordinate sigma-algebra and integrating the independent scalar sum of the complement.

Lemma 6.1 (Gaussian form of the likelihood ratio). *Suppose $\alpha_t \rightarrow \alpha < 1$. Define*

$$A_t(z) = \sqrt{\frac{V}{V_J}} \exp\left(-\frac{z^2}{2V_J}\right). \quad (6.3)$$

Then

$$\mathbb{E}_{Q_t} |R_t(Z_I) - A_t(Z_I)| \rightarrow 0, \quad (6.4)$$

$$\int \mathbb{E}_{Q_t} |H_t(Z_I, d) - A_t(Z_I)| e^{-d} dd \rightarrow 0. \quad (6.5)$$

The functions R_t , H_t , and A_t are bounded by a common finite constant for all sufficiently large t .

Proof. The density upper bound in Theorem 4.3 and Equation (5.2) give the bounds for H_t and R_t ; $A_t \leq \sqrt{V/V_J}$ is also bounded. Moreover, Equation (4.2) gives uniformly in z, d ,

$$\frac{f_{t,J}(-z-d)}{J(t)} = \sqrt{\frac{V}{V_J}} \exp\left(-\frac{(z+d)^2}{2V_J}\right) + o(1).$$

For each fixed D , the displayed Gaussian differs from $A_t(z)$ uniformly in z and $0 \leq d \leq D$ by $O(D/\sqrt{V_J})$. This follows from the bounded derivative of $e^{-u^2/2}$. The part $d > D$ has integral at most Ce^{-D} , uniformly in t . Let $t \rightarrow \infty$ and then $D \rightarrow \infty$ to obtain Equation (6.5). Integrating in d and using Equation (6.1) gives Equation (6.4). $\square$

6.2 Subcritical and positive-fraction blocks

If $\alpha = 0$, then $Z_I/\sqrt{V} \rightarrow 0$ in Q_t -probability, by Chebyshev's inequality. Thus $A_t(Z_I) \rightarrow 1$ in probability. The uniform bounds in Lemma 6.1 imply convergence in L^1 . The likelihood ratio $R_t(Z_I)$ also tends to one in L^1 and is uniformly bounded, giving both total-variation and entropy convergence to zero. For entropy, use continuity and boundedness of $u \log u$ on a fixed interval $[0, C]$, with $0 \log 0 = 0$.

If $0 < \alpha < 1$, the block I is itself macroscopic, so $G_t = Z_I/\sqrt{V_I} \Rightarrow G \sim N(0, 1)$ by Theorem 4.3. Equation (6.3) becomes

$$A_t(Z_I) = \frac{1}{\sqrt{1-\alpha_t}} \exp\left(-\frac{\alpha_t G_t^2}{2(1-\alpha_t)}\right).$$

Hence the likelihood ratio converges in distribution to

$$r_\alpha(G) = \frac{1}{\sqrt{1-\alpha}} \exp\left(-\frac{\alpha G^2}{2(1-\alpha)}\right). \quad (6.6)$$

The common upper bound permits passage to expectations of $|u-1|$ and $u \log u$. Consequently

$$\begin{aligned} \lim d_{\text{TV}}(P_{t,I}, Q_{t,I}) &= \frac{1}{2} \mathbb{E}|r_\alpha(G) - 1|, \\ \lim D_{\text{KL}}(P_{t,I} \| Q_{t,I}) &= \mathbb{E}[r_\alpha(G) \log r_\alpha(G)]. \end{aligned}$$

Now $r_\alpha(g)\varphi(g)$ is the density of $N(0, 1-\alpha)$. It follows directly that

$$\mathbb{E}[r_\alpha(G) \log r_\alpha(G)] = -\frac{1}{2} \log(1-\alpha) - \frac{\alpha}{2}.$$

The inequality $r_\alpha(g) \geq 1$ is equivalent to $|g| \leq \sqrt{1-\alpha} d_\alpha$. Integrating the difference between the $N(0, 1-\alpha)$ and $N(0, 1)$ densities on that interval proves Equation (2.6). Taylor expansion at zero gives Equation (2.9).

For the slack assertion, integrate the absolute difference between Equation (6.2) and Equation (6.1). The triangle inequality and both conclusions of Lemma 6.1 show that

$$\int \mathbb{E}_{Q_t} |H_t(Z_I, d) - R_t(Z_I)| e^{-d} dd \rightarrow 0.$$

This is exactly twice the total variation in Equation (2.7).

6.3 The endpoint fraction one and the converse

Suppose $\alpha_t \rightarrow 1$. The complement J has $V_J = o(V)$, so the already proved subcritical case applies to its coordinates: $d_{\text{TV}}(P_{t,J}, Q_{t,J}) \rightarrow 0$. Under Q_t , Chebyshev gives $Z_J/\sqrt{V} \rightarrow 0$ in probability; the same holds under P_t . By Theorem 5.1, $D_t/\sqrt{V} \rightarrow 0$ under P_t . Since $Z_I = -D_t - Z_J$ under P_t ,

$$Z_I/\sqrt{V} \rightarrow 0 \quad \text{in } P_t\text{-probability.}$$

Under Q_t , the local central limit theorem for I instead gives $Z_I/\sqrt{V} \Rightarrow N(0, 1)$. For a fixed $\varepsilon > 0$, the event $|Z_I| \leq \varepsilon \sqrt{V}$ thus has P_t -probability tending to one and Q_t -probability tending to $q_\varepsilon = 2\Phi(\varepsilon) - 1$. Total variation has lower limit at least $1 - q_\varepsilon$; let $\varepsilon \downarrow 0$.

The data-processing inequality for relative entropy, applied to the indicator of that same event, gives lower limit at least $\log(1/q_\varepsilon)$. This tends to infinity. The inequality itself follows from the log-sum inequality on the event and its complement, so no regularity of the coordinate space is needed.

Finally, if a variance fraction does not tend to zero, some subsequence has a further subsequence converging to an $\alpha \in (0, 1]$. Its total-variation limit is positive and its entropy limit is positive or infinite by the cases just proved. This proves both converse implications in Equation (2.8) and completes Theorem 2.1.

Remark 6.2 (The two Gaussian laws explain the formula). At a positive fraction, the normalized block sum under the product law is asymptotically standard normal. The constraint replaces its variance by $1 - \alpha$. The profile Δ is exactly the total variation between these two centered normal laws. The theorem says that no additional asymptotic discrepancy is hidden in the remaining coordinates of the block: its scalar centered sum asymptotically carries the full likelihood-ratio information.

7 Geometric weights and the sharp Fabius threshold

Fix $q \in (0, 1)$ and set $w_k = q^k$. Write

$$c = \log(1/q), \quad n = n(t) = \left\lfloor \frac{\log t}{c} \right\rfloor, \quad \rho_t = tq^n \in [1, q^{-1}). \quad (7.1)$$

Then $a_k = \rho_t q^{k-n}$. The integer n is the effective dimension, and ρ_t is its residual geometric phase.

Proposition 7.1 (Active dimension). *Uniformly over $\rho_t \in [1, q^{-1})$,*

$$m(t) = n + O_q(1), \quad V(t) = n + O_q(1). \quad (7.2)$$

For every initial segment $I = \{1, \dots, k\}$ with $k \leq n$,

$$V_I = k + O_q(1), \quad (7.3)$$

uniformly in k . If $x = x(t) \downarrow 0$, then

$$n = \frac{\log(1/x) + \log \log(1/x)}{\log(1/q)} + O_q(1). \quad (7.4)$$

Proof. For $k \leq n$, write $j = n - k \geq 0$. Then $a_k = \rho_t q^{-j} \geq q^{-j}$. The defects

$$1 - \mu(a) = \frac{a}{e^a - 1}, \quad 1 - v(a) = \frac{a^2 e^a}{(e^a - 1)^2}$$

are bounded for $a \geq 1$ by constants times $(a + a^2)e^{-a}$. Their sums over the geometrically increasing sequence $\rho_t q^{-j}$ are uniformly finite. This proves the first- n estimates and Equation (7.3).

For $k = n + j$, $j \geq 1$, $a_k = \rho_t q^j$. The mean tail is at most $\frac{1}{2} \sum_{j \geq 1} \rho_t q^j = O_q(1)$, and the variance tail is at most $\frac{1}{4} \sum_{j \geq 1} \rho_t^2 q^{2j} = O_q(1)$. This proves Equation (7.2).

Since $tx = n + O_q(1)$, we have $\log t = \log(1/x) + \log n + O_q(1/n)$. Also $\log t = cn + O_q(1)$. First these imply $n \sim \log(1/x)/c$, then substituting that first-order estimate into $\log n$ gives Equation (7.4). $\square$

Corollary 7.2 (How many Fabius coordinates are independent?). *For $k = k(t) \leq n(t)$ with $k/n \rightarrow \alpha \in [0, 1]$,*

$$d_{\text{TV}}(\mathcal{L}_{P_t}(tq^j U_j : 1 \leq j \leq k), \text{Exp}(1)^{\otimes k}) \rightarrow \Delta(\alpha), \quad (7.5)$$

$$D_{\text{KL}}(\mathcal{L}_{P_t}(tq^j U_j : 1 \leq j \leq k) \parallel \text{Exp}(1)^{\otimes k}) \rightarrow K(\alpha). \quad (7.6)$$

For $k/n \rightarrow 1$, the total variation in Equation (7.5) tends to one and the entropy in Equation (7.6) diverges. Consequently the independent-exponential approximation is valid in total variation if and only if $k = o(n)$, equivalently $k = o(\log(1/x))$.

Proof. The canonical law of the first k scaled coordinates is the product of exponentials conditioned separately to lie in $[0, a_j]$. Its distance from the untruncated product is bounded by

$$\sum_{j \leq k} e^{-a_j} \leq k e^{-a_k}. \quad (7.7)$$

For $k/n \rightarrow \alpha < 1$, $a_k \geq q^{-(n-k)}$ grows exponentially in n , so the bound tends to zero. Equations (2.4) and (7.3) and the triangle inequality give Equation (7.5).

On the truncated box, the canonical density divided by the untruncated exponential density is the constant $\prod_{j \leq k} (1 - e^{-a_j})^{-1}$. Thus

$$D_{\text{KL}}(P_{t,I} \parallel \text{Exp}(1)^{\otimes k}) = D_{\text{KL}}(P_{t,I} \parallel Q_{t,I}) - \sum_{j \leq k} \log(1 - e^{-a_j}),$$

with the understood coordinate scaling. The added term tends to zero for $\alpha < 1$ by Equation (7.7), proving Equation (7.6).

If $k/n \rightarrow 1$, project to the first $\lfloor \beta n \rfloor$ coordinates for any fixed $\beta < 1$. The preceding limits and monotonicity under projection give lower limits $\Delta(\beta)$ and $K(\beta)$; let $\beta \uparrow 1$. The equivalence without a ratio limit follows by the same subsequence argument as in Theorem 2.1. $\square$

For $q = 1/2$, the Laplace product is the transform of the Fabius law. The self-similarity $S_q \stackrel{d}{=} q(U + S'_q)$, with independent U, S'_q , also gives

$$L(t) = \frac{1 - e^{-qt}}{qt} L(qt).$$

In the classical case it yields $F'(x) = 2[F(2x) - F(2x - 1)]$ wherever interpreted through the continuous density, and in particular $F'(x) = 2F(2x)$ for $0 < x < 1/2$. These identities explain why the probabilistic construction is aligned with the repository's functional-equation and transform viewpoints, without importing any unproved repository claim.

8 The infinite boundary law and its geometric phase

The first $o(n)$ coordinates have scaled exponential limits, but the coordinates near index n have widths of order one. Their law remembers ρ_t , even when the bulk limit does not.

Define, for $a > 0$,

$$g_a(u) = \frac{ae^{-au}}{1 - e^{-a}} \mathbf{1}_{[0,1]}(u). \quad (8.1)$$

For an integer j_0 and a phase $\rho \in [1, q^{-1}]$, let

$$\mathcal{B}_{\rho, j_0} = \bigotimes_{j=j_0}^{\infty} g_{\rho q^j}(u_j) du_j. \quad (8.2)$$

This is a probability measure on the countable product $[0, 1]^{\{j_0, j_0+1, \dots\}}$.

Theorem 8.1 (Phase-resolved boundary configuration). *Suppose $w_k = q^k$ and $\rho_t \rightarrow \rho \in [1, q^{-1}]$. For every fixed integer j_0 ,*

$$d_{\text{TV}}(\mathcal{L}_{P_t}((U_{n+j})_{j \geq j_0}, D_t), \mathcal{B}_{\rho, j_0} \otimes \text{Exp}(1)) \rightarrow 0. \quad (8.3)$$

There is no phase-independent limiting law for U_n . The endpoints $\rho = 1$ and $\rho = q^{-1}$ are related by a shift of the boundary index, rather than by equality at the same index.

Proof. The coordinate subset $I_t = \{n + j : j \geq j_0\}$ has total tilted variance $O_{q, j_0}(1)$: it contains at most finitely many indices with $j \leq 0$, and its positive- j variance tail is bounded by a geometric series. Therefore $V_{I_t} = o(V)$, and Theorem 2.1 gives total-variation convergence to the moving product $\mathcal{B}_{\rho_t, j_0}$, jointly with independent exponential slack.

We verify continuity of that *infinite* product in total variation. Let $A(a) = \log \int_0^1 e^{-au} du$. Then $A''(a) = \text{Var}_{g_a}(U) \leq 1/4$. Taylor's theorem for the exponential family gives

$$D_{\text{KL}}(g_a \| g_b) = A(b) - A(a) - (b - a)A'(a) \leq \frac{(a - b)^2}{8}. \quad (8.4)$$

Summing over any finite number of indices and then taking the increasing sigma-algebra limit bounds the infinite-product entropy by

$$D_{\text{KL}}(\mathcal{B}_{\rho, j_0} \| \mathcal{B}_{\rho', j_0}) \leq \frac{(\rho - \rho')^2 q^{2j_0}}{8(1 - q^2)}.$$

The entropy limit can be obtained from finite partitions and the variational characterization of relative entropy. Alternatively, finite-cylinder total-variation bounds pass to the product sigma-algebra by approximation, which already suffices here. Pinsker's inequality consequently gives

$$d_{\text{TV}}(\mathcal{B}_{\rho, j_0}, \mathcal{B}_{\rho', j_0}) \leq \frac{|\rho - \rho'| q^{j_0}}{4\sqrt{1 - q^2}}. \quad (8.5)$$

This proves Equation (8.3).

The mean of the $j = 0$ coordinate is

$$b(\rho) = \frac{1}{\rho} - \frac{1}{e^\rho - 1}. \quad (8.6)$$

Differentiating as an exponential-family mean gives $b'(\rho) = -\text{Var}_{g_\rho}(U) < 0$. Distinct limiting phases therefore produce distinct laws, detected even by their means. Finally $g_{q^{-1}q^j} = g_{q^{j-1}}$, which proves the index-shift statement. $\square$

Corollary 8.2 (The two sides of the transition). *For the one-coordinate laws $g_{\rho q^j}$, as $j \rightarrow -\infty$, the scaled variable $\rho q^j U_j$ converges in total variation to $\text{Exp}(1)$. As $j \rightarrow +\infty$, U_j converges in total variation to $\text{Unif}[0, 1]$.*

Proof. For $a \rightarrow \infty$, h_a is an exponential law conditioned on $[0, a]$, at distance e^{-a} from the unconditioned law. For $a \downarrow 0$, $g_a \rightarrow 1$ uniformly on $[0, 1]$. $\square$

9 A Brownian bridge inside a small Fabius event

Define a random continuous function B_t by its grid values

$$B_t(\ell/n) = \frac{1}{\sqrt{n}} \sum_{k=1}^{\ell} (Y_k - \mu(a_k)), \quad \ell = 0, \dots, n, \quad (9.1)$$

and interpolate linearly. Let B° be a standard Brownian bridge: a centered continuous Gaussian process with covariance $\text{Cov}(B^\circ(s), B^\circ(u)) = \min(s, u) - su$.

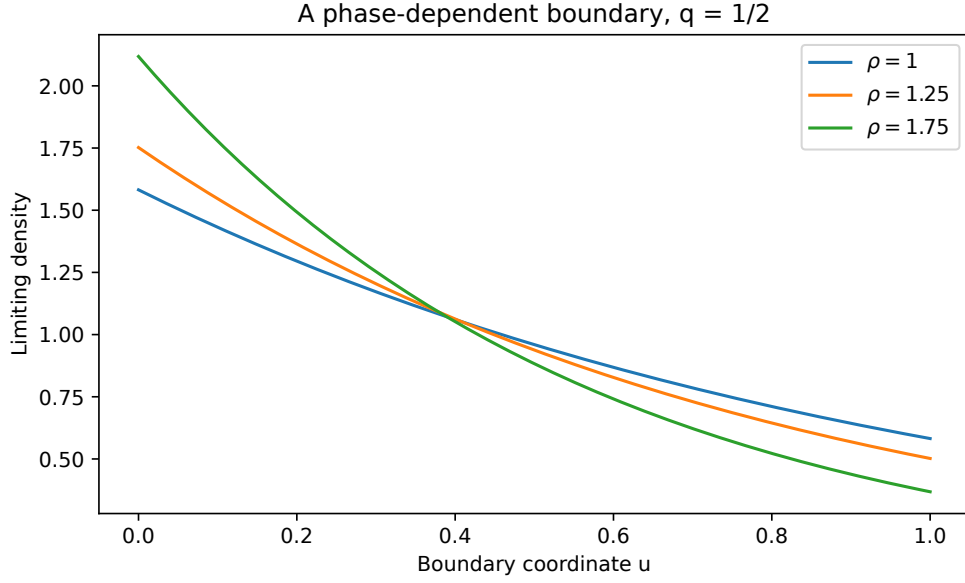


Figure 1: The limiting density of U_n for three geometric phases when $q = 1/2$. This is a phase effect in the *conditional configuration*, not an assertion that the scalar endpoint expansion has a phase-free remainder.

Theorem 9.1 (Joint bridge and slack limit). *For geometric weights,*

$$(B_t, D_t) \Rightarrow (B^\circ, E) \quad \text{in } C[0, 1] \times \mathbb{R}, \quad (9.2)$$

under P_t , where $E \sim \text{Exp}(1)$ is independent of B° . No subsequence restriction on the geometric phase is required.

Proof. Step 1: a product-law invariance principle. Couple the first n tilted coordinates to independent exponentials using independent uniforms V_k :

$$E_k = -\log(1 - V_k), \quad Y_k = -\log(1 - (1 - e^{-a_k})V_k).$$

Then $Y_k \leq E_k$ and $\mathbb{E}(E_k - Y_k) = 1 - \mu(a_k)$. The proof of Proposition 7.1 gives $\sum_{k \leq n} (1 - \mu(a_k)) = O_q(1)$. Therefore the expected supremum norm of the difference between B_t and the linearly interpolated centered exponential walk is at most $O_q(n^{-1/2})$.

For completeness, that exponential walk converges to Brownian motion: independent increment central limit theorems give the finite-dimensional limits, while

$$\mathbb{E} \left| \sum_{k=r+1}^{r+\ell} (E_k - 1) \right|^4 = \ell \mathbb{E}(E_1 - 1)^4 + 3\ell(\ell - 1) = O(\ell^2)$$

gives a useful fourth-moment bound. To check continuous-path tightness explicitly, write an increment of the interpolated walk as $n^{-1/2} \sum c_k (E_k - 1)$, where $0 \leq c_k \leq 1$ and $\sum c_k = n|v - u|$. Independence gives a fourth moment bounded by $Cn^{-2}[(\sum c_k^2)^2 + \sum c_k^4]$. If $|v - u| \geq 1/n$, this is at most $C'|v - u|^2$. If $|v - u| < 1/n$, at most two coefficients are nonzero and their sum is $n|v - u|$, giving the same bound.

For clarity, the resulting dyadic tightness argument is short. Choose $0 < \theta < 1/4$. Markov's inequality and a union bound over all 2^r adjacent dyadic increments at level r bound the probability that any exceeds $A2^{-\theta r}$ by $CA^{-4}2^{-(1-4\theta)r}$. Summing over $r \geq R$ and chaining dyadic approximations controls the modulus of continuity on intervals of length 2^{-R} by a constant times $A2^{-\theta R}$, outside a

probability tending to zero uniformly in n . This proves tightness. Thus under Q_t , $B_t \Rightarrow B$, standard Brownian motion. The same coupling and estimates apply to reversed blocks of the first n coordinates.

Step 2: finite-dimensional conditional distributions. Fix $0 < s_1 < \dots < s_r < 1$ and put $\ell = \lfloor ns_r \rfloor$. Choose $I = \{1, \dots, \ell\}$. The complement has variance $(1 - s_r)n + O_q(1)$. By Equation (6.2) and Lemma 6.1, the joint conditional law of the selected path values and D_t is obtained, up to a bounded likelihood-ratio error tending to zero in L^1 , by weighting the product law by

$$\frac{1}{\sqrt{1 - s_r}} \exp \left(-\frac{B_t(s_r)^2}{2(1 - s_r)} \right) \quad (9.3)$$

and adjoining an independent exponential. The single next coordinate may be included in I without changing its limiting variance fraction. The same bounded likelihood ratio then transfers the vanishing interpolation error from Q_t to P_t .

Under Brownian motion, Equation (9.3) is the likelihood ratio for its restriction to $[0, s_r]$ when conditioned to have value zero at time one. This can be checked directly: the density of the independent remaining increment at $-B(s_r)$, divided by the density of $B(1)$ at zero, is exactly the displayed expression. Gaussian multiplication then gives covariance $\min(s_i, s_j) - s_i s_j$. The extra exponential variable stays independent. This proves all interior finite-dimensional limits.

Step 3: tightness after conditioning. For any coordinate block with complement variance bounded below by a positive fraction of n , Equation (6.1) is uniformly bounded. Apply this once to the first half of the active coordinates, and once to the second half, read in reverse order. Under Q_t both centered half-paths are tight by Step 1. An event with product probability p has conditional probability at most Cp for each such half-path, so both modulus-of-continuity bounds transfer to P_t . Combining the two bounds at the midpoint gives tightness of B_t on $[0, 1]$; its value at the midpoint is tight by the first-half bound.

To identify the endpoint, note that

$$\sqrt{n}B_t(1) = -D_t - Z_{\{k > n\}}.$$

Here $|Z_{\{k > n\}}| \leq \sum_{k > n} a_k = O_q(1)$ deterministically. The slack is tight by Theorem 5.1; hence $B_t(1) \rightarrow 0$ in P_t -probability. Joint tightness with D_t and the identified finite-dimensional distributions prove Equation (9.2). $\square$

This describes two different scales simultaneously. Individual strongly tilted contributions have size t^{-1} . A partial sum of order n such terms fluctuates on scale $\sqrt{n}/t$, while the remaining distance to the inequality boundary is only of order t^{-1} . Pinning the large-scale endpoint produces the bridge; the one-sided microscopic gap produces the independent exponential.

10 Beyond geometric decay

The variance-fraction theorem is not tied to a functional equation or to a dyadic product. It already applies to every positive summable sequence. We calculate its variance clock for two other families, illustrating why the abstract formulation is useful.

10.1 Stretched-exponential weights

Proposition 10.1. *Let $w_k = e^{-ck^\beta}$ with $c, \beta > 0$, and set $N(t) = (\log t/c)^{1/\beta}$. Then*

$$m(t) \sim N(t), \quad V(t) \sim N(t),$$

If $k(t)/N(t) \rightarrow \alpha \in [0, 1]$ and $I_t = \{1, \dots, k(t)\}$, then $V_{I_t}/V(t) \rightarrow \alpha$. Thus Theorem 2.1 gives the same profiles $\Delta(\alpha)$ and $K(\alpha)$ for these prefixes, including the endpoint interpretations.

Proof. Fix $0 < \varepsilon < 1$. For $k \leq (1 - \varepsilon)N$,

$$a_k = \exp\{c(N^\beta - k^\beta)\} \geq \exp\{c[1 - (1 - \varepsilon)^\beta]N^\beta\} \rightarrow \infty$$

uniformly. Their means and variances tend uniformly to one. For $k \geq (1 + \varepsilon)N$, the sums of a_k and a_k^2 tend to zero. To see this explicitly, compare the decreasing function e^{-rcu^β} with its tail integral, for $r = 1, 2$. After $z = rcu^\beta$, the integral is bounded by a polynomial in N times $e^{-rc(1+\varepsilon)^\beta N^\beta}$. Multiplying by $t^r = e^{rcN^\beta}$ still leaves an exponentially small bound. The intermediate indices are at most $2\varepsilon N + O(1)$, each contributing at most one to either mean or variance. Let $\varepsilon \downarrow 0$ to obtain the asymptotics.

The prefix assertion follows from the same three-region argument, with an additional cut at $k(t)$. $\square$

The geometric boundary theorem does not automatically extend in its present form. The transition width in the coordinate index is of order $N^{1-\beta}$ when this grows, rather than uniformly of order one. A boundary scaling theorem for $\beta \neq 1$ is therefore a distinct question, not an unstated corollary.

10.2 Polynomial weights

Proposition 10.2. *Let $w_k = k^{-p}$ with $p > 1$, and let $N = t^{1/p}$. Put*

$$C_p = \int_0^\infty v(u^{-p}) du, \quad M_p = \int_0^\infty \mu(u^{-p}) du. \quad (10.1)$$

Both constants are finite and strictly positive, and

$$V(t) \sim C_p N, \quad m(t) \sim M_p N. \quad (10.2)$$

For a prefix of length $k(t) = \lfloor bN \rfloor$, $0 < b < \infty$, its variance fraction tends to

$$\alpha_p(b) = \frac{\int_0^b v(u^{-p}) du}{C_p} \in (0, 1). \quad (10.3)$$

Its limiting total variation and relative entropy are respectively $\Delta(\alpha_p(b))$ and $K(\alpha_p(b))$.

Proof. Here $a_k = (k/N)^{-p}$. The functions $v(u^{-p})$ and $\mu(u^{-p})$ are continuous on $(0, \infty)$, bounded at zero, and bounded at infinity by constants times u^{-2p} and u^{-p} respectively. They are therefore integrable. Riemann sums on $[\varepsilon, R]$ converge in the usual way. The part $k < \varepsilon N$ contributes at most $\varepsilon N + 1$; the part $k > RN$ is bounded by convergent power tails. Let $\varepsilon \downarrow 0$, $R \rightarrow \infty$. This proves Equation (10.2) and the analogous truncated integral for the prefix. $\square$

Remark 10.3 (A consistency relation between the two constants). The identity $v(a) = \mu(a) - a\mu'(a)$ and integration by parts give $C_p = (1 - 1/p)M_p$. Indeed, for $g(u) = \mu(u^{-p})$, $v(u^{-p}) = g(u) + (u/p)g'(u)$; the boundary term $ug(u)$ vanishes at both endpoints because $p > 1$. This agrees with differentiating $x(t) \sim M_p t^{1/p-1}$ and the exact identity $x'(t) = -V(t)/t^2$, but the integration proof avoids differentiating an asymptotic equivalent without justification.

11 An exact lazy sampler and its expected cost

The original event $S_q \leq x$ is rare, but after tilting the rejection penalty is only of order $n^{-1/2}$. The infinite number of coordinates must nevertheless be handled honestly.

11.1 The ideal accept–reject identity

A proposal from Q_t has total $T = \sum Y_k = tS$. Accept it with probability

$$A(T) = e^{T-m} \mathbf{1}_{T \leq m}. \quad (11.1)$$

This is a valid probability in $[0, 1]$. By Equation (5.1), the accepted distribution is exactly P_t and the acceptance probability is $J(t)$.

Each proposed coordinate can be generated from a scalar uniform V_k by

$$Y_k = -\log(1 - (1 - e^{-a_k})V_k). \quad (11.2)$$

Let H be a separate uniform random variable used for acceptance. For geometric weights, after the first K coordinates have been revealed,

$$s_K = \sum_{k \leq K} Y_k, \quad r_K = \frac{tq^{K+1}}{1-q}, \quad T \in [s_K, s_K + r_K]. \quad (11.3)$$

The bound is deterministic, not a confidence interval.

11.2 Lazy decision rule

Start each proposal by generating the first n coordinates and H . Then apply the following rule, generating one additional coordinate whenever the decision is unresolved.

```

If s_K > m:
    reject.
Else if s_K + r_K <= m:
    If H <= exp(s_K - m):
        accept.
    Else if H > exp(s_K + r_K - m):
        reject.
    Else:
        reveal the next coordinate.
Else:
    reveal the next coordinate.

```

If accepted, the unrevealed coordinates remain available as a lazy continuation. They are not falsely reported as a completed finite representation of an infinite random sequence.

Theorem 11.1 (Exactness and sampling complexity). *In a real-arithmetic model in which $q, t, m(t)$ and the operations in Equation (11.2) are exact, the lazy rule terminates almost surely and its accepted infinite configuration has law P_t . Let K_t be the number of coordinate uniforms drawn for one proposal, excluding H . For geometric weights,*

$$\mathbb{E}K_t = n + O_q(n^{-1/2}), \quad \mathbb{E}[\text{scalar uniform draws per accepted proposal}] \sim \sqrt{2\pi} n^{3/2}. \quad (11.4)$$

The second count includes acceptance uniforms and all rejected proposals; it excludes later requests for additional output coordinates and the cost of precomputing $m(t)$.

Proof. Every decision follows from Equation (11.3) and agrees with the ideal rule $H \leq A(T)$. If $T > m$, eventually $s_K > m$. If $T < m$, eventually $s_K + r_K < m$, and the lower and upper acceptance bounds tend to the common value e^{T-m} . The rule therefore terminates unless $T = m$ or $H = e^{T-m}$. Both events have probability zero: T has a continuous distribution and H is independent and continuous. An accepted finite prefix is valid for every continuation consistent with its bounds. Completing the unrevealed proposal coordinates according to Q_t therefore recovers exactly the ideal accepted infinite law.

For the cost estimate, the first n coordinates have variance $n + O_q(1)$. Their centered sum has density bounded by $C_q/\sqrt{n}$ by Theorem 4.3. Convolution with any additional independent coordinates cannot increase this upper bound. Thus the uncentered sum s_K , for every $K \geq n$, has density bounded by $C_q/\sqrt{n}$.

A proposal unresolved after K lies in one of two disjoint cases. First, $s_K \leq m < s_K + r_K$; its probability is at most $C_q r_K / \sqrt{n}$. Second, $s_K + r_K \leq m$ and H lies strictly between the two exponential acceptance bounds. Its probability is at most

$$\begin{aligned} \frac{C_q}{\sqrt{n}} \int_{-\infty}^{m-r_K} (e^{s+r_K-m} - e^{s-m}) ds &= \frac{C_q}{\sqrt{n}} (1 - e^{-r_K}) \\ &\leq \frac{C_q r_K}{\sqrt{n}}. \end{aligned}$$

Hence $\mathbb{P}(K_t > K) \leq 2C_q r_K / \sqrt{n}$ for $K \geq n$. Since $r_K = \rho_t q^{K-n+1} / (1-q)$, $\sum_{K \geq n} r_K = O_q(1)$. Summing the tail probabilities gives the first assertion of Equation (11.4).

Let $C_t = K_t + 1$ count all scalar uniforms in a proposal. Success has probability $J(t)$, independently across fresh proposal cycles. Although cost and success within a cycle may be dependent, a renewal identity gives expected total cost $\mathbb{E}C_t/J(t)$: condition on the first cycle and restart after failure. Using $J(t) \sim (2\pi V)^{-1/2}$ and $V \sim n$ proves the second assertion. $\square$

Remark 11.2 (Exactness does not mean a proved bit-complexity bound). The theorem counts exact-real coordinate draws. For computable parameters and certified interval access to $m(t)$, the comparisons can be refined until separated; the probability-zero equality cases are the same as above. The series for m has the explicit mean-tail bound $\sum_{k>K} \mu(tq^k) \leq tq^{K+1}/[2(1-q)]$. Nevertheless, arbitrary-precision transcendental evaluation and random-bit costs have not been analyzed here. The accompanying Python experiments use finite binary floating-point arithmetic and a tail cutoff. They are *not* that interval-certified exact implementation.

12 Reproducible checks and numerical experiments

The package contains `code/verify_conditioning.py`, CSV outputs, numerical environment information, and the figure sources in the form of the same reproducible script. No external dataset is needed. All mathematical theorems above are proved independently of this section.

12.1 A finite-simplex model with exact formulas

Let $Q^{(n)}$ be the law of n independent rate-one exponentials, and weight it by $e^{T-n} \mathbf{1}_{T \leq n}$, where T is their sum. The resulting $P^{(n)}$ is uniform on

$$\{(y_1, \dots, y_n) : y_i \geq 0, \sum y_i \leq n\}.$$

This is an exactly solvable surrogate for the strongly tilted active coordinates, not an equality with the geometric series.

For the first $k < n$ coordinates, put $s = \sum_{i \leq k} y_i$. Their likelihood ratio relative to k independent exponentials is

$$R_{n,k}(s) = \frac{n!}{(n-k)!n^k} e^s (1-s/n)^{n-k} \mathbf{1}_{[0,n]}(s). \quad (12.1)$$

Under $P^{(n)}$, s/n has the Beta($k, n+1-k$) law. It follows that

$$\begin{aligned} D_{\text{KL}}(P_k^{(n)} \| Q_k^{(n)}) &= \log \Gamma(n+1) - \log \Gamma(n-k+1) - k \log n \\ &\quad + \frac{nk}{n+1} + (n-k) [\psi(n-k+1) - \psi(n+1)], \end{aligned} \quad (12.2)$$

n	$k = n/2$	TV	KL (nats)	Limiting TV / KL
32	16	0.1652913	0.1052172	0.1660641 / 0.0965736
128	64	0.1658683	0.0988220	0.1660641 / 0.0965736
512	256	0.1660150	0.0971413	0.1660641 / 0.0965736
2048	1024	0.1660518	0.0967159	0.1660641 / 0.0965736

Table 1: The half-dimension finite-simplex comparison. The displayed values evaluate exact formulas; numerical agreement does not itself prove the infinite-series theorem.

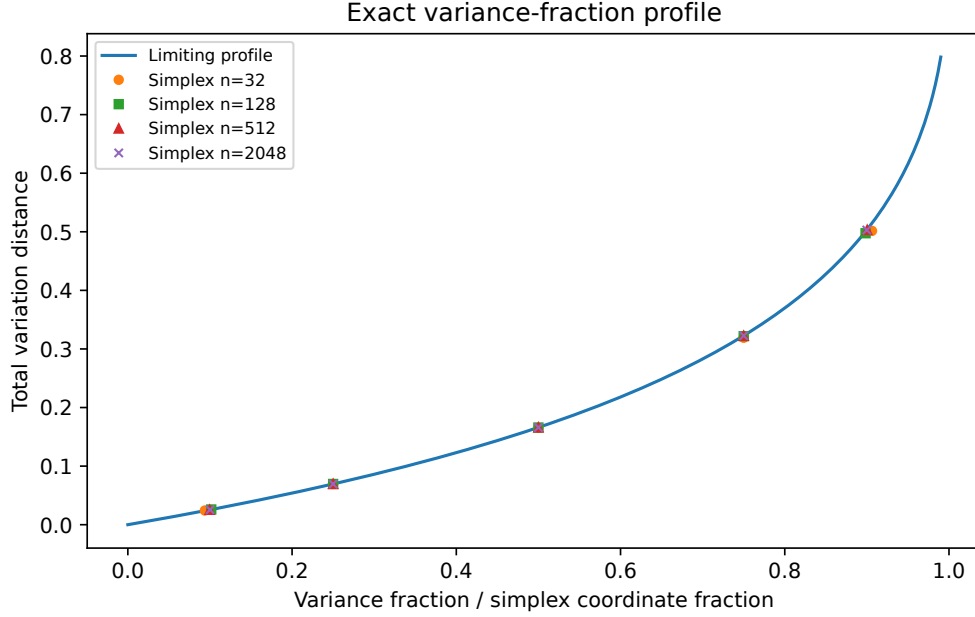


Figure 2: The exact limiting profile Δ , with finite-simplex values from Equation (12.3).

where $\psi = (\log \Gamma)'$.

For total variation, $\log R_{n,k}(s)$ increases up to $s = k$ and then decreases. Let $s_- \leq k \leq s_+$ be its two zeroes, with $s_- = 0$ for $k = 1$. Then

$$d_{\text{TV}}(P_k^{(n)}, Q_k^{(n)}) = P^{(n)}(s_- \leq s \leq s_+) - Q^{(n)}(s_- \leq s \leq s_+), \quad (12.3)$$

which can be evaluated by beta and gamma distribution functions. The formula follows by integrating the positive part of the density difference, so there is no numerical optimization over measurable sets.

12.2 Algebraic and quadrature checks

The script checks the normalization and mean of the truncated exponential, the identity $v = \mu - a\mu'$, and the Taylor coefficients through degree six. It also checks 72 exact polynomial integrals: normalization and first moments for every simplex marginal with $2 \leq n \leq 9$ and $1 \leq k < n$.

Four selected simplex cases are evaluated independently by one-dimensional quadrature. In the delivered run, the maximum absolute discrepancy was below 1.4×10^{-15} for total variation and 3.0×10^{-13} for relative entropy. Stable mean/variance formulas were compared to quadrature over nine widths from 10^{-5} to 60, with maximum absolute discrepancy below 2.3×10^{-16} . These are numerical residuals, not interval-certified error bounds.

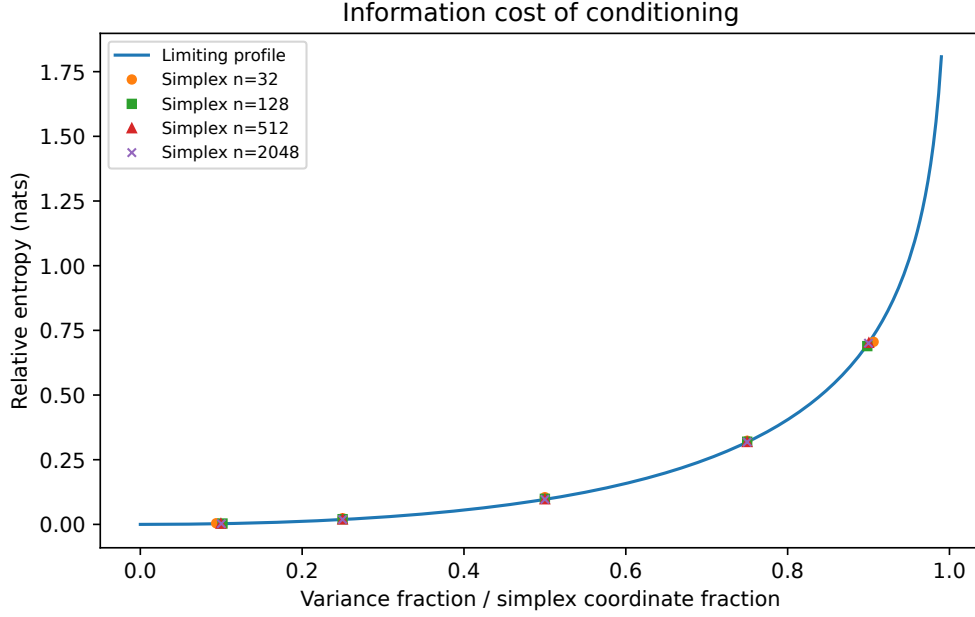


Figure 3: The relative-entropy profile K , with finite-simplex values from Equation (12.2).

n	Accepted	$\widehat{ED}$	SE	$\widehat{\text{Var } B(1/2)}$	$\widehat{EU}_n$
32	11,458	0.9699	0.0089	0.2240	0.4008
128	5,733	0.9965	0.0130	0.2511	0.4025
512	2,733	1.0364	0.0203	0.2507	0.3919

Table 2: Direct tilted-proposal experiments. Limits are 1 for ED, $1/4$ for the midpoint bridge variance, and $1/1.25 - 1/(e^{1.25} - 1) \approx 0.398449$ for EU_n . SE is the empirical standard error of the slack mean. Finite-size bias and sampling fluctuations remain visible.

12.3 Direct geometric-series experiments

For $q = 1/2$, $\rho = 1.25$, and $n = 32, 128, 512$, the script generated 160,000 tilted proposals at each dimension using a fixed documented seed. It retained proposals according to the floating-point version of Equation (11.1). The scaled sum of all omitted support widths was below 2×10^{-13} . This controls deterministic series truncation, not rounding error or the difference between the finite experiment and a certified sampler.

The experiment at $n = 512$ accepted fewer than three thousand configurations, so it should not be read as a high-precision check of all observables. In particular a slack mean above one in that run is compatible with sampling error. The proof establishes the limit; the experiment checks the normalization, indexing, and plausibility of the finite-dimensional formulas.

13 Further research questions and a concrete development plan

The results identify a structure worth extending, rather than closing every issue about conditional Fabius laws. The following questions are proposed research directions, not additional theorems.

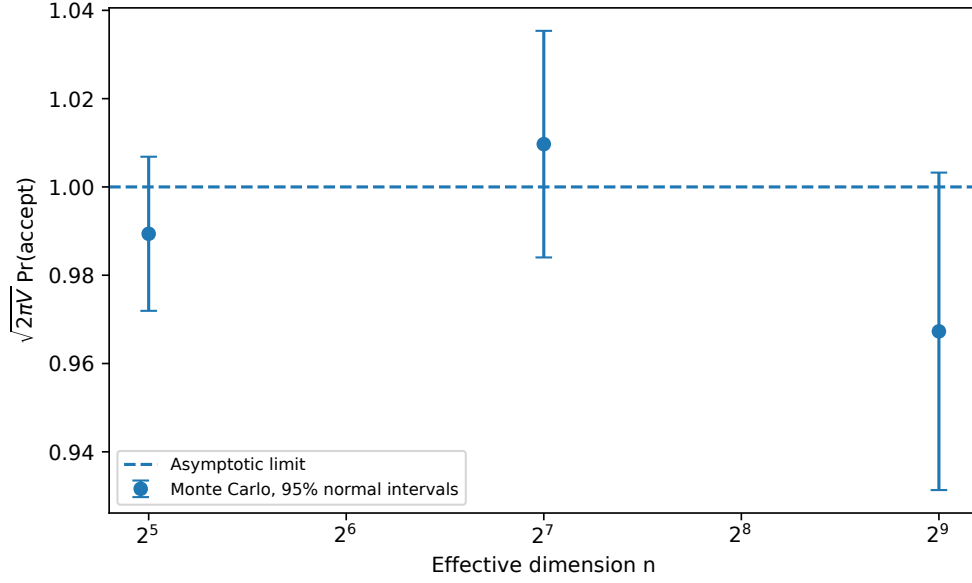


Figure 4: Normalized acceptance probabilities. Bars show approximate 95% normal sampling intervals, not certified enclosures. The predicted limit is one.

13.1 Quantitative and critical-regime questions

Question 13.1 (Uniform finite- t errors). Can one give explicit bounds for the difference between the block likelihood ratio and Equation (6.3), uniform for $\alpha_t \leq 1 - \varepsilon_t$ when $\varepsilon_t \downarrow 0$?

The current proof fixes a positive complement fraction. A useful next target is a bound expressed in the complement variance and its third and fourth cumulants. This would make the threshold usable as a certified finite-precision rule, rather than solely an asymptotic classification.

Question 13.2 (The critical complement). What replaces the Gaussian profile when V_j stays bounded, or tends to infinity much more slowly than V ?

For geometric weights and a block ending within $O(1)$ of n , the complement has the phase-dependent law of Theorem 8.1. Inserting its actual density into Equation (6.1) is a plausible route to a matched macroscopic-to-boundary formula. The theorem that TV tends to one at fraction one does not determine that finer rate.

Editorial note (ProveIt, 2026-09-30). Two independently written packages beside this one answer this question for geometric weights, for different unobserved coordinates; neither cites the other. `.../Critical_Complements_Fabius_Conditioning/` (batch 66 of the repository’s docs/incoming/) hides a fixed number r of early coordinates together with the tail and proves $\sqrt{2\pi n}(1 - d_{TV}) = \Lambda_n + r \log \Lambda_n + \log K - \log(r!) + 1 + o(1)$, $\Lambda_n = \frac{1}{2} \log(2\pi n)$, with the divergences of every fixed Rényi order and an order-zero Rényi crossover; a growing number of hidden early coordinates stays open there. `.../Critical_Complements_Sharp_Information_Loss/` (batch 68) follows the route suggested above: it observes $\{1, \dots, n - m\}$ and hides the last m coordinates before the boundary together with the tail, with the boundary law of Theorem 8.1. For fixed m it gives a two-term overlap expansion with an explicit phase constant and the relative entropy to order $1/n$; for every $m = o(n)$ it gives the leading overlap through the two real branches of Lambert W , with a change of regime at $m \asymp \log n$, and relative entropy $\frac{1}{2} \log(n/m) - \frac{1}{2} + o(1)$ when $m \rightarrow \infty$. The two coincide only when nothing but the tail is hidden, where their corrected overlap theorems agree after the phase change $\rho \mapsto \rho/q$ (the first package’s $\log K = 0.486134172 \dots$ at $q = 1/2$, $\rho = 1$ is the second’s constant $K_{1/2,1/2,0}$). For a

bounded hidden complement both leading overlap terms agree with Equation (5.8), and the second package's full-configuration entropy re-proves Equation (5.7) for geometric weights. General weights and Question 13.3 are not treated.

Question 13.3 (Phase-sensitive higher corrections). Can the repository's all-orders scalar saddle expansions be combined with Equation (6.2) to obtain conditional Edgeworth expansions, including the geometric phase and the slack variable?

This would connect the present configuration-level results directly to the existing exact-phase endpoint machinery. Uniform remainder control in the selected block sum, not just formal Gaussian integration, is the principal analytic obligation.

13.2 Generality and structural extensions

Question 13.4 (Quantitative shape-based local limits). Can the weak-to-uniform upgrade in Lemma 4.2 be made quantitative with a useful universal modulus, and then combined with a Berry–Esseen bound to give explicit conditional error bounds for arbitrary positive summable weights?

The qualitative regularity question is settled here: no weight spacing or nonconcentration assumption is needed. The next issue is effectiveness. The proof near the mode uses both monotonicity and a log-concave midpoint inequality; optimizing that argument may yield a universal, though not necessarily optimal, rate.

Question 13.5 (Several simultaneous constraints). For vector sums $\sum w_k^{(r)} U_k$ constrained near a boundary point, is the block information profile governed by a covariance fraction matrix, with a determinant analogue of $K(\alpha)$?

The scalar Gaussian calculation suggests an expression involving $-\log \det(I-A) - \text{tr } A$, but feasibility cones and multiple slack coordinates change the local geometry. This expression is a heuristic target, not a proved multivariate result.

Question 13.6 (Other endpoint densities and random environments). If the coordinate density behaves as $u^{\gamma-1}$ near zero, can one replace exponentials by gamma limits and obtain the same variance-based profile? For random weights, which quenched hypotheses preserve it, and how does the boundary phase become random?

A proof would need uniform moment and characteristic-function bounds for the new tilted family, together with a precise nonconcentration condition. Merely replacing a density in the final formulas is not enough.

Question 13.7 (Nongeometric boundary fields). What is the limiting transition process for $w_k = e^{-ck^\beta}$, $\beta \neq 1$, and for $w_k = k^{-p}$?

The geometric boundary is a countable product indexed by an $O(1)$ window. A growing transition window suggests a field or a variance-time process rather than the same fixed-index limit. The polynomial fraction $\alpha_p(b)$ supplies a natural clock.

13.3 Algorithms and formal proof

Question 13.8 (Certified random-bit complexity). Can a computable version of the lazy sampler be implemented with interval arithmetic, with proved expected random-bit and arithmetic costs at a requested endpoint precision?

A sensible implementation order is: stable enclosures for μ, v ; certified bounds on m and the geometric tail; a random-bit inverse-CDF primitive; and a terminating comparison loop with a measured precision distribution. Optimality among adaptive proposals is a separate question: Theorem 11.1 proves a cost, not a lower bound on all samplers.

For Lean integration, the most reusable milestones are the truncated-exponential moment identities, the exact scalar likelihood-ratio identity, the abstract local-limit-to-conditioning transfer lemma, and the finite Gaussian integrals producing Δ and K . The shape-based local central limit theorem and the functional tightness argument should be separate modules with explicit hypotheses. Numerical tables must stay outside the trusted proof boundary unless connected by a verified checker, consistent with the repository's stated architecture [7].

14 Conclusion

The passage from endpoint probabilities to conditioned configurations reveals an exact boundary for product behavior. A block is asymptotically independent of the rare-event constraint precisely when it uses a negligible share of the tilted variance. At a positive fraction, its full discrepancy is captured by one Gaussian variance reduction, producing explicit total-variation and entropy profiles. At fraction one the two laws separate.

In the Fabius and geometric cases, that bulk statement coexists with an infinite phase-sensitive transition law, Brownian-bridge partial sums, and independent exponential slack. The same local limit mechanism also yields a lazy exact-real sampler with a polynomial expected draw count in the effective dimension. These are proof-level extensions of the scalar endpoint viewpoint, with quantitative rates, critical complements, and certified bit complexity left as well-defined further problems.

A Additional details for the finite-simplex formulas

The volume of the positive simplex $\{y \in \mathbb{R}_+^n : \sum y_i \leq n\}$ is $n^n/n!$, so its uniform density is $n!/n^n$. Integrating out $n - k$ coordinates leaves

$$p_k(y_1, \dots, y_k) = \frac{n!}{n^n(n-k)!} (n-s)^{n-k} \mathbf{1}_{s \leq n}.$$

Dividing by e^{-s} gives Equation (12.1). The sum of k nonnegative coordinates has cross-sectional factor $s^{k-1}/(k-1)!$, hence density

$$\frac{n!}{(k-1)!(n-k)!n^n} s^{k-1} (n-s)^{n-k}, \quad 0 < s < n.$$

This is the scaled beta density claimed in Section 12. The beta integral and its derivative give

$$\mathbb{E} s = \frac{nk}{n+1}, \quad \mathbb{E} \log(1 - s/n) = \psi(n+1-k) - \psi(n+1),$$

which prove Equation (12.2). Finally

$$\frac{d}{ds} \log R_{n,k}(s) = 1 - \frac{n-k}{n-s} = \frac{k-s}{n-s}.$$

For $k > 1$, $R_{n,k}(0) = \prod_{j=0}^{k-1} (1 - j/n) < 1$; at its maximum it must exceed one because it integrates to one under Q_k and is not constant. Thus it has exactly the two crossings used in Equation (12.3). For $k = 1$, the lower crossing is zero. These facts justify the root brackets in the numerical script.

The finite model also reproduces the profile by elementary asymptotics. Under the product law, $(s - k)/\sqrt{k} \Rightarrow N(0, 1)$. Taylor expansion of $\log R_{n,k}(s)$ around $s = k$ and Stirling's formula, for $k/n \rightarrow \alpha \in (0, 1)$, give

$$\log R_{n,k}(s) = -\frac{1}{2} \log(1 - \alpha) - \frac{\alpha}{2(1 - \alpha)} \left(\frac{s - k}{\sqrt{k}} \right)^2 + o(1)$$

on central windows. This is the same r_α as in Equation (6.6); the infinite-series proof supplies the missing uniform likelihood-ratio control directly, rather than inferring it from this finite analogy.

B Proof dependency and repository integration audit

Result	Main inputs	Status in this package
Tilt and moment formulas	Product integration, bounded differentiation	Proved here
Subset density theorem	Uniform moments, log-concavity, shape convergence	Proved here
Slack and full entropy	Density theorem, exact change of measure	Proved here
Sharp block profiles	Complement density, Gaussian integrals	Proved here
Geometric boundary	Subcritical-block theorem, product continuity	Proved here
Brownian bridge	Product invariance principle, bounded density ratios	Proved here
Other weight families	Riemann sums and elementary tail estimates	Proved here
Lazy sampler	Exact acceptance identity, geometric tail bounds	Proved in stated model
Python diagnostics	Symbolic checks and floating-point calculations	Executed, not formal proofs
Lean declarations	None supplied or compiled for these new results	Not claimed

The repository was consulted at commit `e2b1f016a94102663f12b970e6dd434229f90d01`. The principal specific source read was

```
Analysis/FabiusFunction/Lean/FabiusFunction/PaperFabiusAsymptotic.lean
```

whose blob is `d9508bee558197a19a90b615f9ee84e29ee3b793`. Its claim-level audit explains the status of the scalar endpoint expansions. The frontier README and draft manifest were also consulted to understand the mixed formalized/prose research boundary. This was a targeted source inspection, not a repository build or an exhaustive claim-by-claim comparison with its entire research corpus.

No theorem above is labeled Lean-verified merely because nearby scalar results have Lean counterparts. Conversely, the proofs here do not assume the correctness of every claim in the repository. All distributional assertions needed for the new theorem package are established in this article from the displayed random-series model with positive summable weights.

C Reproduction instructions and artifact scope

The source file `article.tex` compiles with pdfLaTeX and standard TeX Live packages, including Libertinus. The figures and two table fragments are included in the package, so rebuilding the PDF does not require rerunning Monte Carlo.

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

To rerun the diagnostic suite and regenerate the data and figures:

```
python -m pip install -r requirements.txt
python code/verify_conditioning.py --proposals 160000
python code/make_tables.py
```

The first command is optional when dependencies are already installed. The environment used for the delivered run is recorded in `data/environment.json`. The script uses a documented fixed pseudorandom seed; exact byte-for-byte agreement of plots across library versions is not promised. No font files are included.

References

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- [3] P. Diaconis and D. A. Freedman, *Conditional limit theorems for exponential families and finite versions of de Finetti’s theorem*, Journal of Theoretical Probability **1** (1988), 381–410. doi:10.1007/BF01048727.
- [4] A. Dembo and O. Zeitouni, *Refinements of the Gibbs conditioning principle*, Probability Theory and Related Fields **104** (1996), 1–14. doi:10.1007/BF01303799.
- [5] E. Cator and P. A. Ferrari, *Gibbs conditioning principle for log-concave independent random variables*, arXiv:2512.24910v2, 21 April 2026. <https://arxiv.org/abs/2512.24910v2>. The cited comparison is to the explicit integer-valued setup in this version.
- [6] A. Prékopa, *On logarithmic concave measures and functions*, Acta Scientiarum Mathematicarum (Szeged) **34** (1973), 335–343. The one-dimensional convolution argument used here is reproved in Lemma 4.1.
- [7] V. Reshetnikov and contributors, *ProveIt*, repository snapshot e2b1f016a94102663f12b970e6dd434229f90d01, consulted 28 September 2026 (Pacific time). <https://github.com/VladimirReshetnikov/ProveIt/tree/e2b1f016a94102663f12b970e6dd434229f90d01>. In particular, the Fabius endpoint claim audit at `Analysis/FabiusFunction/Lean/FabiusFunction/PaperFabiusAsymptotic.lean`.